Practical 15: Notifications with NotificationManager

Source: MODULE 2.pdf | Source pages: 44–49

Aim

To develop and test an Android application demonstrating notifications with notificationmanager.

Steps to Perform the Practical

1. Create an Android Studio project for the notification practical.

2. Add a button to the main layout to trigger a notification.

3. Follow the supplied code to create a notification channel where required by the Android version.

4. Build a Notification using NotificationCompat and display it with NotificationManager.

5. Add any required notification permission in AndroidManifest.xml and request it at runtime if the device requires it.

6. Run the app, tap the button, and check the notification shade for the notification.

Result

The application is run and the expected behavior is verified according to the practical instructions and code.

Note: Use the package name and resource names from the source code consistently. If Android Studio reports a missing resource or class, check spelling and ensure all files are in the correct folders.

8. Develop an Android application implementing Notifications using Notification Manager.

Project Name: NotificationDemo

File Name: activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>

<LinearLayout

   xmlns:android="http://schemas.android.com/apk/res/android"

   android:layout_width="match_parent"

   android:layout_height="match_parent"

   android:gravity="center">

   <Button

       android:id="@+id/btnNotify"

       android:layout_width="wrap_content"

       android:layout_height="wrap_content"

       android:text="Show Notification"/>

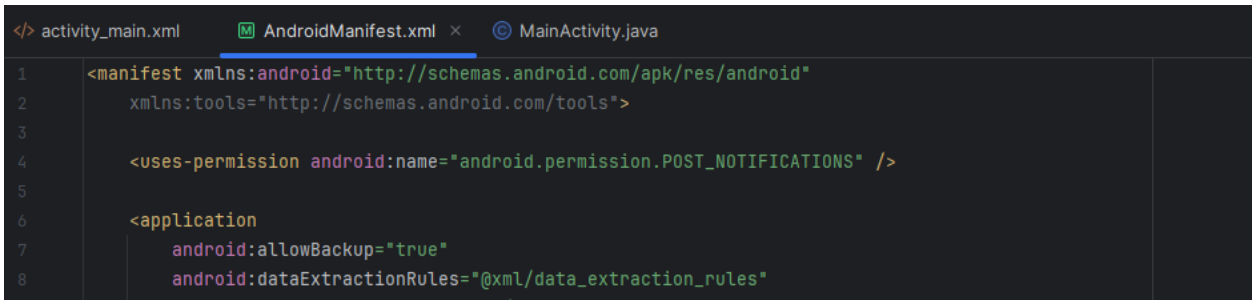
</LinearLayout>
```

Add Permission in Manifest,

Open: AndroidManifest.xml

Add above <application>:

<uses-permission android:name="android.permission.POST_NOTIFICATIONS”/>



File Name: MainActivity.java

```
package com.example.notificationdemo;


import android.Manifest;

import android.app.NotificationChannel;

import android.app.NotificationManager;

import android.content.pm.PackageManager;

import android.os.Build;

import android.os.Bundle;

import android.widget.Button;

import androidx.appcompat.app.AppCompatActivity;

import androidx.core.app.NotificationCompat;


public class MainActivity extends AppCompatActivity {


   private static final String CHANNEL_ID = "MyChannel";

   private static final int NOTIFICATION_PERMISSION_CODE = 100;


   private Button btnNotify;


   @Override

   protected void onCreate(Bundle savedInstanceState) {

       super.onCreate(savedInstanceState);

       setContentView(R.layout.activity_main);
```

```
        // Find the button from activity_main.xml
        btnNotify = findViewById(R.id.btnNotify);

        // Create notification channel
        createNotificationChannel();

        // Request notification permission for Android 13+
        requestNotificationPermission();

        // Button click
        btnNotify.setOnClickListener(v -> {
            showNotification();
        });
    }

    // Request notification permission
    private void requestNotificationPermission() {

        if (Build.VERSION.SDK_INT >= Build.VERSION_CODES.TIRAMISU) {

            if (checkSelfPermission(Manifest.permission.POST_NOTIFICATIONS)
                    != PackageManager.PERMISSION_GRANTED) {

                requestPermissions(
                        new String[]{Manifest.permission.POST_NOTIFICATIONS},
                        NOTIFICATION_PERMISSION_CODE
```

```
            );
        }
    }
}

// Create notification channel
private void createNotificationChannel() {

    if (Build.VERSION.SDK_INT >= Build.VERSION_CODES.O) {

        NotificationChannel channel = new NotificationChannel(
                CHANNEL_ID,
                "My Channel",
                NotificationManager.IMPORTANCE_DEFAULT
        );

        channel.setDescription("Notification Channel for Demo");

        NotificationManager manager =
                getSystemService(NotificationManager.class);

        if (manager != null) {
            manager.createNotificationChannel(channel);
        }
    }
}
```

```
// Show notification
private void showNotification() {

    // Check permission for Android 13+
    if (Build.VERSION.SDK_INT >= Build.VERSION_CODES.TIRAMISU) {

        if (checkSelfPermission(Manifest.permission.POST_NOTIFICATIONS)
                != PackageManager.PERMISSION_GRANTED) {

            requestNotificationPermission();
            return;
        }
    }

    NotificationCompat.Builder builder =
            new NotificationCompat.Builder(this, CHANNEL_ID)
                    .setSmallIcon(android.R.drawable.ic_dialog_info)
                    .setContentTitle("Notification Example")
                    .setContentText("Hello from Android Notification")
                    .setPriority(NotificationCompat.PRIORITY_DEFAULT)
                    .setAutoCancel(true);

    NotificationManager manager =
            (NotificationManager) getSystemService(NOTIFICATION_SERVICE);
```

```
        if (manager != null) {

            manager.notify(1, builder.build());

        }

    }

}
```

